

A self-consistent bootstrap for the book algorithm and the bound $R(k, k) \leq 3.769^{k+o(k)}$

[author list to be completed]*

Abstract

Gupta, Ndiaye, Norin and Wei [4] recast the Campos–Griffiths–Morris–Sahasrabudhe book algorithm [2] as an inductive statement, obtaining $R(k, k) \leq (3.7992\dots)^{k+o(k)}$ by two rounds of a bootstrap in which an already established bound on off-diagonal Ramsey numbers enlarges the set of admissible parameters of the next round; they remark that further rounds are possible. We show that the bound *being proved* may itself be used to define the admissible set, so that the whole (infinite) iteration collapses to the verification of a single system of inequalities. Optimising within this framework—no longer over a closed-form ansatz but over free-form rate functions, discretised as concave piecewise-quadratic splines—gives

$$R(k, \ell) \leq e^{F(\ell/k)k+o(k)} \quad (1 \leq \ell \leq k)$$

for an explicit such F , presented as an exact-integer certificate of 35 770 cells with one tangent witness per cell and orientation, with $e^{F(1)} \leq 3.7689719$; in particular $R(k, k) \leq 3.769^{k+o(k)}$. Self-consistency also removes the hand analysis of small ratios: below $\lambda_0 = 10^{-4}$ the certified F is simply extended linearly, at the cost of a single inequality. The entire proof has been formalised in Lean 4 (Remark 4.4): the final theorems, including the fully formal decimal bound $R(k, k) \leq 3.7690^k$ for all large k , depend only on the axioms `propext`, `Classical.choice`, `Quot.sound`. We also describe how the admissible region must be computed in both orientations $\ell \leq k$ and $\ell > k$ (the point corrected between the first and second arXiv versions of [4]), and note that a proposed quintic correction with constant 3.6961 [5] does not satisfy the self-consistency inequality.

1 Introduction

Let $R(k, \ell)$ be the least N such that every red/blue colouring of the edges of K_N contains a red K_k or a blue K_ℓ . Erdős and Szekeres [3] proved $R(k, \ell) \leq \binom{k+\ell-2}{k-1}$, so $R(k, k) \leq 4^k$. After decades of lower-order improvements, Campos, Griffiths, Morris and Sahasrabudhe [2] proved $R(k, k) \leq (4 - 2^{-7})^k$ with their *book algorithm*. Gupta, Ndiaye, Norin and Wei [4] replaced the algorithm by an inductive statement, optimised the parameters and obtained, for $\ell \leq k$,

$$R(k, \ell) \leq e^{G(\ell/k)k+o(k)} \binom{k+\ell}{\ell}, \quad G(\lambda) = (-0.25\lambda + 0.03\lambda^2 + 0.08\lambda^3)e^{-\lambda}, \quad (1)$$

which at $\lambda = 1$ gives $R(k, k) \leq (4e^{-0.14/e})^{k+o(k)} = (3.7992027\dots)^{k+o(k)}$.

Throughout, $H(\lambda) = (1 + \lambda) \log(1 + \lambda) - \lambda \log \lambda$ for $\lambda > 0$, so that $\binom{k+\ell}{\ell} \leq e^{H(\ell/k)k}$ is the Erdős–Szekeres bound in exponential form.

*Revision v3, 2 September 2026. The combinatorial core is the book algorithm of Campos, Griffiths, Morris and Sahasrabudhe [2] in the inductive form of Gupta, Ndiaye, Norin and Wei [4]; Section 3 reproves it, with explicit hypotheses, for the reader’s convenience.

Theorem 1.1. *There is a concave, strictly increasing, piecewise-quadratic C^1 function $F: (0, 1] \rightarrow (0, \infty)$, given explicitly by the certificate of Section 5 (a spline on a partition of $[\lambda_0, 1]$, $\lambda_0 = 10^{-4}$, into 10 560 intervals, extended linearly on $(0, \lambda_0]$), with the following property: for every $\varepsilon > 0$ there is N_ε such that*

$$R(k, \ell) \leq \exp((F(\ell/k) + \varepsilon)k) \quad \text{for all integers } k \geq \ell \geq 1 \text{ with } k + \ell \geq N_\varepsilon.$$

Its terminal value is the exact rational

$$F(1) = \frac{2653604512524788171208898}{2 \cdot 10^{24}} = 1.32680225\dots, \quad e^{F(1)} \leq 3.7689719 < 3.769;$$

in particular

$$R(k, k) \leq 3.769^{k+o(k)}.$$

λ	0.1	0.25	0.5	0.75	0.9	1
$H(\lambda)$	0.3351	0.6255	0.9548	1.1951	1.3143	1.3863
$H(\lambda) + G(\lambda)$ [4]	0.3128	0.5793	0.8896	1.1304	1.2565	1.3348
$F(\lambda)$ (Theorem 1.1)	0.3023	0.5680	0.8823	1.1230	1.2487	1.3268

Table 1: Exponents in $R(k, \lambda k) \leq e^{(\cdot)^{k+o(k)}}$. The values of F are evaluated from the certificate nodes (Section 5.2).

The improvement over (1) at the diagonal is the factor $(3.76897/3.79920)^k = e^{-0.00799k}$; Table 1 lists some off-diagonal values.

Context and what is new

The proof in [4] has two parts. The combinatorial part (their Lemma 12 and Theorem 13) says, roughly, that a candidate pair (X, Y) with red density $\geq p$ and $|X||Y| \geq x^{-k}y^{-\ell}\mu^{-\ell}$ contains one of the desired cliques, provided $x < (1 - \mu)p^{1/(1-\mu)}$ and (x, y) is *admissible*, i.e. $x^{-k'}y^{-\ell'}$ is (asymptotically) an upper bound for $R(k', \ell')$ for all k', ℓ' . The analytic part (their Theorem 14) converts this into a bound $R(k, \ell) \leq e^{F(\ell/k)k+o(k)}$ for any smooth F such that, with $p = 1 - e^{-F'}$ and $X = (1 - M)p^{1/(1-M)}$, some admissible $(X(\lambda), Y(\lambda))$ satisfies

$$F(\lambda) > -\frac{1}{2}(\log X(\lambda) + \lambda \log M(\lambda) + \lambda \log Y(\lambda)) \quad (0 < \lambda \leq 1). \quad (2)$$

The larger the admissible $Y(\lambda)$, the smaller F can be; and the admissible set is enlarged by any already established bound on Ramsey numbers. In [4] this is exploited in two rounds: the Erdős–Szekeres bound gives a first $F_{0.08}$, which enlarges the admissible set and gives $F_{0.03}$, i.e. (1). The authors remark that additional rounds are possible and report that an unverified third round would give 3.78233 [4, Remark 17].

Our contribution is threefold.

1. *Self-consistency* (Section 4). We prove that the admissible set may be computed from the very function F one is trying to establish (Theorem 4.2). A direct induction is circular at its base; we resolve this with a *shift ladder*: the functions $F + s$, $s \downarrow 0$, are shown valid one after another, each step being an application of the one-round theorem to the previous one. This replaces the infinite iteration by a single system of inequalities and yields the

fixed point of the bootstrap. The ladder theorem—and, as of this revision, the entire proof of Theorem 1.1—is formalised in Lean 4 on top of the RamseyLean development of [4] (Remark 4.4).

2. *Optimisation and certification* (Section 5). Instead of optimising within a closed-form ansatz (as the previous revision of this paper did, with a degree-8 correction to the Erdős–Szekeres exponent, reaching 3.772), we compute the fixed point of the bootstrap in a discretisation of the framework: F is a free-form concave piecewise-quadratic spline on a 10 560-interval partition of $[\lambda_0, 1]$, $\lambda_0 = 10^{-4}$, determined interval by interval by the minimal admissible slope, with a per-interval book parameter M and pair (X, Y) , admissibility being tested against F itself—self-consistently, and in both orientations. The maximal admissible Y is controlled by a Legendre-type transform of the symmetric extension of F (Lemma 5.1, equivalent to [4, Lemma 15]), so that one tangent-line witness per cell and orientation suffices. The proof object is a certificate of 35 770 cells whose entries are exact scaled integers; self-consistency permits a *linear* extension of F below λ_0 , in place of the hand analysis of small ratios that a closed-form ansatz requires. The certificate is verified by a Python checker and, bit-identically, by the Lean kernel.
3. *Both orientations* (Section 6). The admissible region must dominate the Ramsey bound at all ratios $\ell/k \in (0, \infty)$; the first arXiv version of [4] used the bound only for $\ell \leq k$, which its second version corrects. We indicate the precise point, give a numerical witness, and note that the corrected, self-consistent framework confirms the constant 3.7992... directly and rules out a proposed quintic with constant 3.6961 [5].

2 Notation, valid bounds and admissible pairs

All colourings are red/blue edge colourings of a complete graph. $N_R(v)$ and $N_B(v)$ are the red and blue neighbourhoods of a vertex v . For disjoint non-empty vertex sets X, Y , $e_R(X, Y)$ is the number of red edges between them and $d(X, Y) = e_R(X, Y)/(|X||Y|)$. A *candidate* is a pair (X, Y) of disjoint non-empty sets; it is (k, ℓ, t) -good if $X \cup Y$ contains a red K_k , or X contains a blue K_t , or Y contains a blue K_ℓ . As K_1 is a vertex, every candidate is (k, ℓ, t) -good when $k = 1$ or $t = 1$. A *blue book* (S, T) consists of disjoint S, T with S a blue clique and all S – T edges blue.

Definition 2.1 (valid functions). $G: (0, 1] \rightarrow \mathbb{R}$ is *valid* if for every $\varepsilon > 0$ there is N such that $R(k, \ell) \leq \exp((G(\ell/k) + \varepsilon)k)$ for all integers $k \geq \ell \geq 1$ with $k + \ell \geq N$. Its *symmetric extension* is $\tilde{G}(t) = G(t)$ for $0 < t \leq 1$ and $\tilde{G}(t) = tG(1/t)$ for $t > 1$.

By $R(k, \ell) = R(\ell, k)$, if G is valid then for all $k, \ell \geq 1$ with $k + \ell \geq N$, writing $t = \ell/k$,

$$R(k, \ell) \leq \exp(\tilde{G}(t)k + \varepsilon \max(k, \ell)) \quad (3)$$

(for $\ell > k$: $R(\ell, k) \leq e^{(G(k/\ell) + \varepsilon)\ell}$ and $G(k/\ell)\ell = \tilde{G}(t)k$). The function H is valid with $N = 1$, $\varepsilon = 0$: $\binom{k+\ell}{\ell} \leq (k+\ell)^{k+\ell} k^{-k} \ell^{-\ell} = e^{H(\ell/k)k}$.

Definition 2.2 (admissible pairs). For $G: (0, 1] \rightarrow \mathbb{R}$, a pair $(x, y) \in (0, 1)^2$ is *G-admissible* if $-\log x - t \log y \geq \tilde{G}(t)$ for all $t > 0$.

Thus (x, y) is *G-admissible* iff $x^{-k}y^{-\ell} \geq e^{\tilde{G}(\ell/k)k}$ for all $k, \ell \geq 1$, i.e. iff $x^{-k}y^{-\ell}$ dominates the bound (3) (up to the o -term) for *every* ratio, in both orientations. The set $\{(x, y) : G\text{-admissible}\}$ is closed under decreasing x or y .

3 The key lemma

Lemma 3.4 is [4, Lemma 12] (a compression of the book algorithm of [2]) with the hypotheses on Ramsey numbers made explicit. We include the complete proof, following [4].

Lemma 3.1 (Cauchy–Schwarz). *For every candidate (X, Y) : $\sum_{v \in X} e_R(X, N_R(v) \cap Y) = \sum_{y \in Y} \deg_R(y, X)^2 \geq e_R(X, Y)^2/|Y|$.*

Proof. Both sums count triples (v, u, y) with $v, u \in X$, $y \in Y$, vy and uy red; then apply Cauchy–Schwarz to $\sum_y \deg_R(y, X)$. $\square$

Lemma 3.2 (books). *Let $0 < \mu < 1$, let $b, m, k \geq 1$ be integers with $m \geq 5b^2/\mu$, and let X be a vertex set with $|X| \geq 5m^2$ such that at least $R(k, m)$ vertices $v \in X$ satisfy $|N_B(v) \cap X| \geq \mu|X|$. Then X contains a red K_k or a blue book (S, T) with $|S| = b$ and $|T| \geq \frac{1}{2}\mu^b|X|$.*

Proof. Let $W = \{v \in X : |N_B(v) \cap X| \geq \mu|X|\}$. Since $|W| \geq R(k, m)$, W contains a red K_k (done) or a blue K_m with vertex set U . For $z \in X \setminus U$ let $\sigma_z = |N_B(z) \cap U|/m$. Then $m \sum_{z \in X \setminus U} \sigma_z = e_B(U, X \setminus U) \geq \sum_{u \in U} (|N_B(u) \cap X| - (m-1)) \geq m(\mu|X| - m)$, so the mean $\bar{\sigma}$ of the σ_z satisfies $\bar{\sigma} \geq \mu - m/|X|$. Pick $S \subseteq U$ uniformly at random with $|S| = b$ and let $T = \{z \in X \setminus U : S \subseteq N_B(z)\}$, so that (S, T) is a blue book. If $\sigma_z m \geq b$ then

$$\Pr[z \in T] = \binom{\sigma_z m}{b} / \binom{m}{b} = \prod_{i=0}^{b-1} \frac{\sigma_z m - i}{m - i} \geq \left(\frac{\sigma_z m - b}{m - b} \right)^b \geq \left(\sigma_z - \frac{b}{m} \right)_+^b,$$

since $(a - i)/(m - i)$ is non-increasing in i when $a \leq m$. By convexity of $u \mapsto (u - b/m)_+^b$,

$$\mathbb{E}|T| \geq \sum_{z \in X \setminus U} \left(\sigma_z - \frac{b}{m} \right)_+^b \geq |X \setminus U| \left(\bar{\sigma} - \frac{b}{m} \right)_+^b \geq (|X| - m) \left(\mu - \frac{m}{|X|} - \frac{b}{m} \right)_+^b.$$

Here $b/m \leq \mu/(5b)$ and $m/|X| \leq 1/(5m) \leq \mu/(25b^2) \leq \mu/(25b)$, so the last factor is at least $\mu^b(1 - \frac{6}{25b})^b \geq 0.76\mu^b$ (Bernoulli), and $|X| - m \geq (1 - \frac{1}{5m})|X| \geq 0.8|X|$. Hence $\mathbb{E}|T| \geq 0.608\mu^b|X|$. $\square$

Lemma 3.3. *Let $0 < \eta < 1/2$, $p, \mu \in [\eta, 1 - \eta]$ and let $r \geq 2\eta^{-1} \log(1/\eta)$ be an integer. Then*

$$(p^{1/r} - \mu)^r (1 - \mu)^{1-r} \geq (1 - \mu) p^{1/(1-\mu)} \left(1 - \frac{\log^2(1/\eta)}{\eta^2 r} \right).$$

Proof. Let $u = \log p/(1 - \mu)$, so $|u| \leq \eta^{-1} \log(1/\eta) \leq r/2$. From $p^{1/r} \geq 1 + (\log p)/r$ we get $(p^{1/r} - \mu)/(1 - \mu) \geq 1 + u/r > 0$. For $|v| \leq 1/2$, $\log(1 + v) \geq v - v^2$, so $(1 + u/r)^r \geq e^{u - u^2/r} \geq e^u(1 - u^2/r)$. Multiply by $1 - \mu$. $\square$

Lemma 3.4 (key lemma). *Let $0 < \eta < 1/4$, $0 < \varepsilon < \eta/4$, let $r \geq 1$ be an integer and let $x, y, \mu, p \in (0, 1)$ satisfy $\mu \geq \eta$, $p - \varepsilon \geq \eta$, $x + \varepsilon < 1$, $y + \varepsilon < 1$, $\mu + \varepsilon < 1$ and*

$$x^{1/r} (1 - \mu - \varepsilon)^{1-1/r} + \mu + 2\varepsilon \leq (p - \varepsilon)^{1/r}, \quad (4)$$

$$\varepsilon(\mu + x) \leq \mu(1 - \mu - x). \quad (5)$$

There is $L_0 = L_0(\varepsilon, r, \eta)$ such that for all integers $\ell \geq L_0$, $k \geq 1$, $t \geq 1$ the following holds. Suppose

(H1) $R(k', \ell) \leq (x + \varepsilon)^{-k'}(y + \varepsilon)^{-\ell}$ for every integer $1 \leq k' \leq k$;

(H2) (X, Y) is a candidate such that, with $\delta_s := \varepsilon/s$ and $\alpha_0 := d(X, Y) + \delta_{k+t} - p$,

$$\alpha_0 > 0 \quad \text{and} \quad \alpha_0^r |X| |Y| \geq x^{-k} y^{-\ell} \mu^{-t}. \quad (6)$$

Then (X, Y) is (k, ℓ, t) -good.

Choice of L_0 . Write $n = k + t$, $N = k + \ell + t$ and

$$b = \left\lceil \frac{r \log((1 + \varepsilon)n^2/\varepsilon) + \log 2}{\log(1 + \varepsilon/\mu)} \right\rceil, \quad m = \lceil 5b^2/\mu \rceil, \quad w = (n + m)^m. \quad (7)$$

Since $\mu \geq \eta$, we have $\log(1 + \varepsilon/\mu) \geq \log(1 + \varepsilon)$ and $m \leq \lceil 5b^2/\eta \rceil$, so $b = O(\log n)$, $m = O(\log^2 n)$ and $\log(2n^3w/(\varepsilon\eta)) = O(\log^3 n)$ with constants depending only on ε, r, η . Hence there are K_1, C_1 depending only on (ε, r, η) such that $(1 + \varepsilon)^n \geq \max\{5m^2, 2n^3w/(\varepsilon\eta), 1/\varepsilon\}$ for $n \geq K_1$ and $\max\{\dots\} \leq C_1$ for $n < K_1$. Let $(1 + \varepsilon)^{L_0-r} \geq C_1$. Then, for $\ell \geq L_0$ and all $k, t \geq 1$ (using $p - \varepsilon \geq \eta$),

$$(1 + \varepsilon)^{N-r} \geq \max\left\{5m^2, \frac{2n^3w}{\varepsilon(p - \varepsilon)}, \frac{1}{\varepsilon}\right\}. \quad (8)$$

□

Proof of Lemma 3.4. Fix $\ell \geq L_0$ and induct on $n = k + t$; the cases $k = 1, t = 1$ are trivial, so let $k, t \geq 2$ and assume the statement for smaller $k + t$ (same ℓ and parameters).

Step 0. If $|Y| \geq (x + \varepsilon)^{-k}(y + \varepsilon)^{-\ell}$ then $|Y| \geq R(k, \ell)$ by (H1), so Y contains a red K_k or a blue K_ℓ and we are done. Hence assume $|Y| < (x + \varepsilon)^{-k}(y + \varepsilon)^{-\ell}$. As $\alpha_0 \leq 1 + \varepsilon$, (6) gives

$$|X| \geq \frac{x^{-k} y^{-\ell} \mu^{-t}}{\alpha_0^r |Y|} > \frac{(1 + \varepsilon/x)^k (1 + \varepsilon/y)^\ell \mu^{-t}}{(1 + \varepsilon)^r} \geq (1 + \varepsilon)^{N-r}, \quad (9)$$

and also $\alpha_0 |X| \geq (1 + \varepsilon)^{N-r}$ (if $\alpha_0 \leq 1$ then $\alpha_0 |X| \geq \alpha_0^r |X| > (1 + \varepsilon)^N$ by the same computation; if $\alpha_0 > 1$ then $\alpha_0 |X| > |X|$). By (8),

$$|X| \geq 5m^2, \quad w \leq \frac{\varepsilon(p - \varepsilon)}{2n^3} |X|, \quad \frac{1}{\alpha_0 |X|} < \varepsilon. \quad (10)$$

Step 1: regularisation. If some $v \in X$ has $|N_R(v) \cap Y| < (p - \delta_n)|Y| = (d(X, Y) - \alpha_0)|Y|$, put $X' = X \setminus \{v\}$. Then $e_R(X', Y) > d(X, Y)|X||Y| - (d(X, Y) - \alpha_0)|Y|$, so $d(X', Y) > d(X, Y) + \alpha_0/(|X| - 1)$ and $\alpha'_0 := d(X', Y) + \delta_n - p > \alpha_0 |X|/(|X| - 1)$, whence $(\alpha'_0)^r |X'| |Y| > \alpha_0^r |X| |Y|$. Thus (X', Y) satisfies (H2) (and $X' \neq \emptyset$), and goodness of (X', Y) implies that of (X, Y) . Repeating, we may assume

$$|N_R(v) \cap Y| \geq (p - \delta_n)|Y| \geq (p - \varepsilon)|Y| \quad (v \in X); \quad (11)$$

(9)–(10) remain valid since they only used (6) and $|Y|$.

Step 2: many vertices of large blue degree. Let $W = \{v \in X : |N_B(v) \cap X| \geq (\mu + \varepsilon)|X|\}$ and note $R(k, m) \leq \binom{k+m}{m} \leq w$. Suppose $|W| \geq w$. By Lemma 3.2 (with $\mu + \varepsilon$ for μ ; $m \geq 5b^2/(\mu + \varepsilon)$ and $|X| \geq 5m^2$), X contains a red K_k (done) or a blue book (S, T) with $|S| = b$, $|T| \geq \frac{1}{2}(\mu + \varepsilon)^b |X|$.

If $t \leq b$ then $S \supseteq$ a blue K_t (done). Otherwise put $t' = t - b \geq 1$ and consider (T, Y) with parameters (k, ℓ, t') . By (11), $d(T, Y) \geq p - \delta_n$, so

$$\alpha_0(T, Y) := d(T, Y) + \delta_{k+t'} - p \geq \delta_{n-b} - \delta_n = \frac{\varepsilon b}{n(n-b)} \geq \frac{\varepsilon}{n^2} > 0,$$

and by (6), $\alpha_0 \leq 1 + \varepsilon$ and the choice (7) of b ,

$$\alpha_0(T, Y)^r |T| |Y| \geq \left(\frac{\varepsilon}{n^2}\right)^r \frac{(\mu + \varepsilon)^b}{2} \cdot \frac{x^{-k} y^{-\ell} \mu^{-t}}{(1 + \varepsilon)^r} = \frac{(1 + \varepsilon/\mu)^b}{2} \left(\frac{\varepsilon}{(1 + \varepsilon)n^2}\right)^r x^{-k} y^{-\ell} \mu^{-t'} \geq x^{-k} y^{-\ell} \mu^{-t'}.$$

(H1) is unchanged. By induction (T, Y) is (k, ℓ, t') -good; a blue $K_{t'}$ in T together with S is a blue K_t in X , and the other outcomes are inherited. So (X, Y) is good.

Step 3: choosing a vertex. Now let $|W| < w$. For $v \in X$ let $Y_v = N_R(v) \cap Y \neq \emptyset$ and $d_v = d(X, Y_v)$; put $d = d(X, Y)$. Lemma 3.1 gives $\sum_v |Y_v| d_v \geq d^2 |X| |Y| = d \sum_v |Y_v|$, i.e. $\sum_{v \in X} |Y_v| (d_v - d) \geq 0$; the terms with $v \in W$ contribute less than $w|Y|$. If $d_v < d - \gamma$ for all $v \in X \setminus W$, where $\gamma = \varepsilon/n^3$, then by (11) $-w|Y| < -\gamma(|X| - w)(p - \varepsilon)|Y|$, so $w > \gamma(p - \varepsilon)|X|/2$, contradicting (10). Fix therefore $v \in X \setminus W$ with

$$d(X, Y') \geq d(X, Y) - \frac{\varepsilon}{n^3}, \quad |Y'| \geq (p - \varepsilon)|Y|, \quad |N_B(v) \cap X| < (\mu + \varepsilon)|X|, \quad Y' := N_R(v) \cap Y, \quad (12)$$

and let $X_R = N_R(v) \cap X$, $X_B = N_B(v) \cap X$, $p' = p - \delta_{n-1}$, $\alpha = d(X, Y') - p'$, $\alpha_R = d(X_R, Y') - p'$, $\alpha_B = d(X_B, Y') - p'$ (the last two when the sets are non-empty). Since $\delta_{n-1} - \delta_n = \varepsilon/(n(n-1)) \geq \varepsilon/n^3$, (12) gives $\alpha \geq \alpha_0 > 0$.

Step 4a: red step. Suppose $X_R \neq \emptyset$, $\alpha_R > 0$ and $\alpha_R^r |X_R| \geq \frac{x}{p - \varepsilon} \alpha^r |X|$. Consider (X_R, Y') with parameters $(k - 1, \ell, t)$: (H1) holds a fortiori, $\alpha_0(X_R, Y') = \alpha_R > 0$, and

$$\alpha_R^r |X_R| |Y'| \geq \frac{x}{p - \varepsilon} \alpha^r |X| (p - \varepsilon) |Y| \geq x \alpha_0^r |X| |Y| \geq x^{-(k-1)} y^{-\ell} \mu^{-t}.$$

By induction (X_R, Y') is $(k - 1, \ell, t)$ -good; as v is red to all of $X_R \cup Y'$, a red K_{k-1} there gives a red K_k , and the other outcomes are inherited. So (X, Y) is good.

Step 4b: blue step. Suppose $X_B \neq \emptyset$, $\alpha_B > 0$ and $\alpha_B^r |X_B| \geq \frac{\mu}{p - \varepsilon} \alpha^r |X|$. Consider (X_B, Y') with parameters $(k, \ell, t - 1)$: (H1) holds, $\alpha_0(X_B, Y') = \alpha_B > 0$ and $\alpha_B^r |X_B| |Y'| \geq \mu \alpha_0^r |X| |Y| \geq x^{-k} y^{-\ell} \mu^{-(t-1)}$. By induction it is $(k, \ell, t - 1)$ -good; a blue K_{t-1} in X_B with v is a blue K_t in X . So (X, Y) is good.

Step 4c: neither applies—contradiction. Splitting $e_R(X, Y') - p' |X| |Y'|$ along $X = X_R \sqcup X_B \sqcup \{v\}$ and using $e_R(\{v\}, Y') = |Y'|$ gives $\alpha |X| |Y'| = \alpha_R |X_R| |Y'| + \alpha_B |X_B| |Y'| + (1 - p') |Y'|$ (terms for empty sets being 0), hence

$$1 \leq \frac{(\alpha_R |X_R|)_+}{\alpha |X|} + \frac{(\alpha_B |X_B|)_+}{\alpha |X|} + \frac{1}{\alpha |X|}. \quad (13)$$

As Step 4a does not apply, either $(\alpha_R |X_R|)_+ = 0$ or $\alpha_R |X_R| = (\alpha_R^r |X_R|)^{1/r} |X_R|^{1-1/r} < (\frac{x}{p - \varepsilon})^{1/r} \alpha |X|^{1/r} |X_R|^{1-1/r}$; similarly for B . With $a = |X_B|/|X| \in [0, \mu + \varepsilon)$ and $|X_R|/|X| \leq 1 - a$, (13) yields

$$(p - \varepsilon)^{1/r} \leq f(a) + \frac{(p - \varepsilon)^{1/r}}{\alpha |X|}, \quad f(a) := x^{1/r} (1 - a)^{1-1/r} + \mu^{1/r} a^{1-1/r}. \quad (14)$$

For $r \geq 2$, $f'(a) = (1 - \frac{1}{r})[(\mu/a)^{1/r} - (x/(1-a))^{1/r}] \geq 0$ iff $a \leq \mu/(\mu+x)$, and (5) says $\mu + \varepsilon \leq \mu/(\mu+x)$; so f is non-decreasing on $[0, \mu + \varepsilon]$ (constant if $r = 1$). Using $\mu^{1/r}(\mu + \varepsilon)^{1-1/r} \leq \mu + \varepsilon$ (weighted AM-GM), $f(a) \leq f(\mu + \varepsilon) \leq x^{1/r}(1 - \mu - \varepsilon)^{1-1/r} + \mu + \varepsilon$. Finally $\alpha|X| \geq \alpha_0|X| > 1/\varepsilon$ by (10), so (14) gives $(p - \varepsilon)^{1/r} < x^{1/r}(1 - \mu - \varepsilon)^{1-1/r} + \mu + 2\varepsilon$, contradicting (4). $\square$

4 The bootstrap

4.1 One round

The following is [4, Theorem 14] with two harmless modifications: the identity defining X is relaxed to an inequality (the proof only ever uses the inequality), and admissibility is phrased through a valid function G rather than through the set $\mathcal{R}$ of [4]. We include the proof to make all constants explicit.

Theorem 4.1 (one round). *Let $G: (0, 1] \rightarrow \mathbb{R}$ be valid. Let $F: (0, 1] \rightarrow [0, \infty)$ be C^1 with $F' > 0$ and $C_F := \sup_{0 < \lambda \leq 1} (H(\lambda) - F(\lambda))/\lambda < \infty$, and let $M, X, Y: (0, 1] \rightarrow (0, 1)$ be continuous with*

$$X(\lambda) \leq (1 - M(\lambda))p(\lambda)^{1/(1-M(\lambda))}, \quad p(\lambda) := 1 - e^{-F'(\lambda)}. \quad (15)$$

Assume that for every $\lambda \in (0, 1]$:

(A1) $(X(\lambda), Y(\lambda))$ is G -admissible;

(A2) $\psi(\lambda) := F(\lambda) + \frac{1}{2}(\log X(\lambda) + \lambda \log M(\lambda) + \lambda \log Y(\lambda)) > 0$.

Then F is valid.

Proof. Fix $\varepsilon \in (0, 1/4]$. We shall find K such that $R(k, \ell) \leq e^{(F(\ell/k) + 2\varepsilon)k}$ for all $k \geq K$ and $1 \leq \ell \leq k$; since $k \geq (k + \ell)/2$, this gives validity with $N = 2K$.

Small ratios. Put $\varepsilon_0 = \min\{1, \varepsilon/(2C_F)\}$ if $C_F > 0$ and $\varepsilon_0 = 1$ otherwise. For $\ell \leq \varepsilon_0 k$, $R(k, \ell) \leq e^{H(\lambda)k} \leq e^{(F(\lambda) + C_F \lambda)k} \leq e^{(F(\lambda) + \varepsilon/2)k}$, $\lambda = \ell/k$. Let $I = [\varepsilon_0, 1]$.

Constants. By continuity and compactness choose $\eta \in (0, 1/4)$ with $X, Y, M, p \in [\eta, 1 - \eta]$ on I , let $\psi_{\min} = \min_I \psi > 0$, and put $\theta = \min\{2\varepsilon, \psi_{\min}/2\} \leq 1/2$. By uniform continuity of F' on I choose $\theta' \in (0, \min_I F'/2)$ so that $p_\lambda := 1 - e^{-F'(\lambda) + \theta'}$ satisfies $p_\lambda \geq \eta/2$ and $\log(p(\lambda)/p_\lambda) \leq \theta\eta/4$ on I . Define

$$x_\lambda = X(\lambda)e^{-\theta}, \quad y_\lambda = Y(\lambda)e^{-\theta}, \quad \mu_\lambda = M(\lambda) \quad (\lambda \in I),$$

so $x_\lambda, y_\lambda \in [\eta/2, 1 - \eta]$. By (A1),

$$-\log x_\lambda - t \log y_\lambda \geq \tilde{G}(t) + \theta(1 + t) \quad (t > 0), \quad (16)$$

and by (15), since $1/(1 - \mu_\lambda) \leq 1/\eta$,

$$x_\lambda \leq (1 - \mu_\lambda)p_\lambda^{1/(1-\mu_\lambda)} \exp\left(\frac{\log(p(\lambda)/p_\lambda)}{1 - \mu_\lambda} - \theta\right) \leq (1 - \mu_\lambda)p_\lambda^{1/(1-\mu_\lambda)} e^{-3\theta/4}.$$

By Lemma 3.3 (with $\eta/2$) fix an integer $r \geq 2$ with $(p_\lambda^{1/r} - \mu_\lambda)^r (1 - \mu_\lambda)^{1-r} \geq (1 - \mu_\lambda)p_\lambda^{1/(1-\mu_\lambda)} e^{-\theta/4}$ for all $\lambda \in I$. Then

$$x_\lambda^{1/r} (1 - \mu_\lambda)^{1-1/r} \leq e^{-\theta/(2r)} (p_\lambda^{1/r} - \mu_\lambda) \quad (\lambda \in I), \quad (17)$$

so in particular $p_\lambda^{1/r} - \mu_\lambda \geq x_\lambda^{1/r}(1 - \mu_\lambda)^{1-1/r} \geq \eta/2$. Now fix $\varepsilon_1 > 0$ with

$$(a) \ 2\varepsilon_1 + \frac{4\varepsilon_1}{\eta} \leq (1 - e^{-\theta/(2r)})\frac{\eta}{2}, \quad (b) \ \varepsilon_1 \leq \frac{\eta^3}{4}, \quad (c) \ \varepsilon_1 \leq \frac{\theta\eta}{8}, \quad (d) \ \varepsilon_1 \leq \frac{\eta}{32}.$$

We check the hypotheses of Lemma 3.4 for $(x_\lambda, y_\lambda, \mu_\lambda, p_\lambda)$ with ε_1 in place of ε and $\eta/4$ in place of η , uniformly in $\lambda \in I$. Clearly $\varepsilon_1 < \eta/16$, $\mu_\lambda \geq \eta/4$, $p_\lambda - \varepsilon_1 \geq \eta/4$, and $x_\lambda + \varepsilon_1, y_\lambda + \varepsilon_1, \mu_\lambda + \varepsilon_1 < 1$. For (4), by (17) and (a),

$$x_\lambda^{1/r}(1 - \mu_\lambda - \varepsilon_1)^{1-1/r} + \mu_\lambda + 2\varepsilon_1 \leq p_\lambda^{1/r} - (1 - e^{-\theta/(2r)})\frac{\eta}{2} + 2\varepsilon_1 \leq p_\lambda^{1/r} - \frac{4\varepsilon_1}{\eta} \leq (p_\lambda - \varepsilon_1)^{1/r},$$

the last step because $\frac{d}{du}u^{1/r} = \frac{1}{r}u^{1/r-1} \leq u^{-1} \leq 4/\eta$ on $[\eta/4, 1]$. For (5): $p < 1$ and $1/(1-M) \geq 1$ give $X \leq (1-M)p$, so $1 - \mu_\lambda - x_\lambda \geq (1-M)(1-p) \geq \eta^2$ and $\varepsilon_1(\mu_\lambda + x_\lambda) \leq 2\varepsilon_1 \leq \eta^3/2 \leq \mu_\lambda(1 - \mu_\lambda - x_\lambda)$ by (b). Let $L_0 = L_0(\varepsilon_1, r, \eta/4)$ be as in Lemma 3.4.

The Ramsey bounds consumed by (H1). Let N_G be such that (3) holds for G with $\theta/2$ in place of ε whenever $k + \ell \geq N_G$, and let $L_1 = \max\{L_0, N_G\}$. Let $\ell \geq L_1$, $\lambda \in I$ and $k' \geq 1$ be arbitrary, and $t = \ell/k'$. By (c), $\log(x_\lambda + \varepsilon_1) \leq \log x_\lambda + \varepsilon_1/x_\lambda \leq \log x_\lambda + \theta/4$, and likewise for y_λ ; with (16),

$$-k' \log(x_\lambda + \varepsilon_1) - \ell \log(y_\lambda + \varepsilon_1) \geq k'(\tilde{G}(t) + \frac{3}{4}\theta(1+t)) \geq \tilde{G}(t)k' + \frac{\theta}{2} \max(k', \ell) \geq \log R(k', \ell),$$

where the last inequality is (3) ($k' + \ell \geq N_G$). Thus (H1) holds for every k .

The induction. Let $K \geq \max\{L_1/\varepsilon_0, 2/\varepsilon_0\}$ be so large that for all $k \geq K$: (i) $|F'(s) - F'(s')| \leq \theta'/2$ whenever $s, s' \in [\varepsilon_0/2, 1]$, $|s - s'| \leq 1/k$; (ii) $e^{-\varepsilon k} \leq 1 - e^{-\theta'/2}$ (recall $F \geq 0$); (iii) $2\varepsilon k \geq r \log(2k/\varepsilon_1) + \log 16$; (iv) $e^{(F(\varepsilon_0) + \varepsilon)k} \geq 4$; (v) $e^{\varepsilon k} \geq 2$. Fix $k \geq K$. We prove by induction on $\ell \in \{1, \dots, k\}$ that every colouring of K_N with $N := \lceil e^{(F(\ell/k) + \varepsilon)k} \rceil$ contains a red K_k or a blue K_ℓ . For $\ell \leq \varepsilon_0 k$ this was shown above. Let $\ell > \varepsilon_0 k$, $\lambda = \ell/k \in I$, and consider a colouring of K_N .

Low red density. If the red density is $< p_\lambda$, some vertex has blue degree at least $(1 - p_\lambda)(N - 1) = e^{-F'(\lambda) + \theta'}(N - 1) \geq e^{(F(\lambda) + \varepsilon)k - F'(\lambda) + \theta'/2}$ by (ii). By the mean value theorem, $F(\frac{\ell-1}{k})k = F(\lambda)k - F'(\xi)$ with $\xi \in [\lambda - \frac{1}{k}, \lambda] \subseteq [\varepsilon_0/2, 1]$, and $F'(\xi) \geq F'(\lambda) - \theta'/2$ by (i). Hence the blue neighbourhood has at least $\lceil e^{(F((\ell-1)/k) + \varepsilon)k} \rceil$ vertices and, by the induction hypothesis (or the small-ratio case), contains a red K_k or a blue $K_{\ell-1}$, which with the vertex gives a blue K_ℓ .

High red density. If the red density is $\geq p_\lambda$, choose disjoint X, Y with $|X| = |Y| = \lfloor N/2 \rfloor$ maximising $e_R(X, Y)$. For a uniformly random such pair, each edge lies between X and Y with probability $|X||Y|/\binom{N}{2}$, so $\mathbb{E} d(X, Y)$ equals the red density and $d(X, Y) \geq p_\lambda$. Apply Lemma 3.4 with parameters $(x_\lambda, y_\lambda, \mu_\lambda, p_\lambda, \varepsilon_1, r)$, this $\ell \geq \varepsilon_0 k \geq L_1$, and $t = \ell$. (H1) was verified. For (H2): $\alpha_0 = d(X, Y) + \varepsilon_1/(k + \ell) - p_\lambda \geq \varepsilon_1/(2k) > 0$, and by the definitions of x_λ, y_λ , the definition of ψ , and $\theta(1 + \lambda) \leq 2\theta \leq \psi_{\min} \leq 2\psi(\lambda)$,

$$x_\lambda^{-k} y_\lambda^{-\ell} \mu_\lambda^{-\ell} = \exp(k[-\log X - \lambda \log Y - \lambda \log M + \theta(1 + \lambda)]) = \exp(k[2F(\lambda) - 2\psi(\lambda) + \theta(1 + \lambda)]) \leq e^{2F(\lambda)k},$$

while $\alpha_0^r |X||Y| \geq (\varepsilon_1/2k)^r (N/2 - 1)^2 \geq (\varepsilon_1/2k)^r e^{2(F(\lambda) + \varepsilon)k}/16 \geq e^{2F(\lambda)k}$ by (iv) and (iii). Lemma 3.4 shows that (X, Y) is (k, ℓ, ℓ) -good, i.e. the colouring contains a red K_k or a blue K_ℓ .

Consequently $R(k, \ell) \leq e^{(F(\ell/k) + \varepsilon)k} + 1 \leq e^{(F(\ell/k) + 2\varepsilon)k}$ for all $k \geq K$, $\ell \leq k$, by (v). $\square$

4.2 Self-consistency

Theorem 4.2 (self-consistent bootstrap). *Let $F: (0, 1] \rightarrow [0, \infty)$ be C^1 with $F' > 0$ and $C_F < \infty$, and let $M, X, Y: (0, 1] \rightarrow (0, 1)$ be continuous with $X(\lambda) = (1 - M(\lambda))p(\lambda)^{1/(1-M(\lambda))}$, $p = 1 - e^{-F'}$, such that for all $\lambda \in (0, 1]$:*

(B1) $(X(\lambda), Y(\lambda))$ is F -admissible: $-\log X(\lambda) - t \log Y(\lambda) \geq \tilde{F}(t)$ for all $t > 0$;

(B2) $\psi(\lambda) = F(\lambda) + \frac{1}{2}(\log X(\lambda) + \lambda \log M(\lambda) + \lambda \log Y(\lambda)) > 0$.

Then F is valid.

Proof. For $\sigma \geq 0$ put $F_\sigma(\lambda) = F(\lambda) + \sigma(1 + \lambda)$, $X_\sigma = X e^{-\sigma}$, $Y_\sigma = Y e^{-\sigma}$. We check the hypotheses of Theorem 4.1 for the function $F_{\sigma/2}$, the valid input $G = F_\sigma$ and the triple (M, X_σ, Y_σ) .

- $F_{\sigma/2}$ is C^1 , ≥ 0 , $C_{F_{\sigma/2}} \leq C_F$, and $F'_{\sigma/2} = F' + \sigma/2 > 0$, so $p_{\sigma/2} := 1 - e^{-F' - \sigma/2} \geq p$ and $X_\sigma \leq X = (1 - M)p^{1/(1-M)} \leq (1 - M)p_{\sigma/2}^{1/(1-M)}$, which is (15).
- The symmetric extension of F_σ is $\tilde{F}_\sigma(t) = \tilde{F}(t) + \sigma(1 + t)$ for all $t > 0$ (for $t > 1$: $t F_\sigma(1/t) = t F(1/t) + \sigma(t + 1)$). Hence, by (B1), $-\log X_\sigma - t \log Y_\sigma = -\log X - t \log Y + \sigma(1 + t) \geq \tilde{F}_\sigma(t)$: the pair (X_σ, Y_σ) is F_σ -admissible, which is (A1).
- (A2) for $F_{\sigma/2}$ with (X_σ, Y_σ) reads $F + \frac{\sigma}{2}(1 + \lambda) + \frac{1}{2}(\log X - \sigma + \lambda \log M + \lambda \log Y - \lambda \sigma) = \psi(\lambda) > 0$.

Thus, by Theorem 4.1, if F_σ is valid then $F_{\sigma/2}$ is valid. Since $F \geq 0$, $F_{\log 4} \geq \log 4$, and the constant $\log 4$ is valid ($R(k, \ell) \leq \binom{k+\ell-2}{k-1} \leq 4^k$); so $F_{\log 4}$ is valid, and by induction so is F_{σ_j} with $\sigma_j = 2^{-j} \log 4$ for every j . Given $\varepsilon > 0$ choose j with $2\sigma_j \leq \varepsilon/2$; as $F_{\sigma_j} \leq F + \varepsilon/2$ on $(0, 1]$, validity of F_{σ_j} with $\varepsilon/2$ gives validity of F with ε . $\square$

Remark 4.3. Theorem 4.2 will be applied, in Section 5, to data that are only piecewise continuous. Its proof (through Theorem 4.1) uses the continuity of M, X, Y and of ψ only to produce, on each compact $I = [\varepsilon_0, 1]$, uniform bounds $X, Y, M, p \in [\eta, 1 - \eta]$ and $\inf_I \psi > 0$, and it uses the identity for X only through the inequality (15). The theorem therefore holds verbatim with “continuous” replaced by these two conclusions—in particular for the piecewise-quadratic C^1 spline F of Section 5, whose X and Y are step functions with finitely many values and whose per-cell slack bounds are part of the certificate. The formalised ladder theorem (Remark 4.4) is stated in exactly this weakened form and is applied unchanged. Self-consistency is also what legitimises the tail of the certificate: below λ_0 the input bound against which admissibility is tested is F itself, not the Erdős–Szekeres bound, so F may simply be continued by a straight line (Section 5.2).

Remark 4.4. Theorem 4.2 has been formalised in Lean 4 on top of the RamseyLean development accompanying [4] (which formalises Lemma 3.4 and Theorem 4.1 as `Candidate.isGood_of_density_card_produ` and `uniformRamseyExpBound_of_descent`): file `SelfConsistent.lean`, theorem `uniformRamseyExpBound_self` depending only on the axioms `propext`, `Classical.choice`, `Quot.sound`. The only change to the existing development is the relaxation of the identity for X in their Theorem 14 to the inequality (15), whose proof already used only the inequality. The proof above is the formalised one; the shift $\sigma(1 + \lambda)$ (rather than a constant) is what makes every rung need only the pointwise slack $\psi > 0$.

Since the first version of this paper, the *entire* proof of Theorem 1.1 has been formalised; for the present revision the development is the directory `RamseyLean/Bootstrap2` of the artefact. The rate function is *defined from the certificate data*: D and M are the piecewise-linear interpolants of the nodes (clamped outside $[\lambda_0, 1]$) and $F := F_0 + \int_{\lambda_0}^\cdot D$, so that `HasDerivAt F (D r) r` is the fundamental theorem of calculus, and the node values $F(\lambda_j) = F_j$ are exact by the integer chain identity (19). A Bool-valued cell checker (integer interval arithmetic at scale 10^{12} , `exp/log` enclosed by Taylor series with directed rounding), mirrored bit-exactly by the Python verifier of Section 5.3, is proved sound against the real-valued definitions; the validity of the tangent witnesses (each is literally a node of the certificate) is established by a monotone two-pointer merge pass; and the ladder theorem of the preceding paragraph is applied unchanged (Remark 4.3). Every cell is discharged by the Lean *kernel* (`decide +kernel`; `native_decide` is not used): 35 770 cells at 0.3–0.6 s per cell (Apple M-series / AMD Genoa), about six core-hours over 559 chunk files of 64 cells each; a checking process needs about 1.5 GB, mostly the import closure (against 5.3 GB per 48-cell chunk for the ansatz certificate of the previous revision). The final theorems, each depending only on the axioms `propext`, `Classical.choice`, `Quot.sound`, are `main_bootstrap2` (validity of the spline rate function), `bootstrap2_diagonal` ($R(k, k) \leq e^{(F_N + \varepsilon)k}$ eventually, for every $\varepsilon > 0$, with the explicit rational F_N of Theorem 1.1), and `bootstrap2_diagonal_num`: $R(k, k) \leq (37690/10000)^k$ for all large k —a fully formal decimal bound, where the previous revision stated the decimal form only in prose.

Remark 4.5. The ladder is what makes it legitimate to feed the bound F into its own admissible region. A direct induction on $k + \ell$ seems to fail: the region hypothesis is needed for pairs (k', ℓ) with $k' < k$ and the same ℓ , whose sums may fall below any fixed starting point of the induction, and the trivial bound $R \leq 4^{k+\ell}$ at the base is exponentially too weak to be absorbed.

5 The maximal admissible Y and the spline certificate

5.1 A Legendre description of the admissible boundary

For fixed x , the largest y with (x, y) F -admissible is $y = e^{-S}$ with $S = \sup_{t>0} (\tilde{F}(t) + \log x)/t$. When $\tilde{F}$ is concave the supremum is located by a tangent line; this is the content of [4, Lemma 15], in the following form.

Lemma 5.1. *Let $\varphi: (0, \infty) \rightarrow \mathbb{R}$ be concave, $u < 0$, and $S(u) = \sup_{t>0} (\varphi(t) + u)/t$. If $t_0 > 0$ and σ is a supergradient of φ at t_0 (i.e. $\varphi(t) \leq \varphi(t_0) + \sigma(t - t_0)$ for all t) with $\varphi(t_0) - \sigma t_0 + u \leq 0$, then $S(u) \leq \sigma$.*

Proof. $(\varphi(t) + u)/t \leq (\varphi(t_0) - \sigma t_0 + u)/t + \sigma \leq \sigma$ for every $t > 0$. □

Suppose $F: (0, 1] \rightarrow (0, \infty)$ is concave and C^1 with $F' > 0$, and let $M, X: (0, 1] \rightarrow (0, 1)$ be given. Put

$$S(\lambda) = \sup_{t>0} \frac{\tilde{F}(t) + \log X(\lambda)}{t}, \quad Y(\lambda) = e^{-S(\lambda)}, \quad \psi(\lambda) = F(\lambda) + \frac{1}{2}(\log X + \lambda \log M - \lambda S)(\lambda), \quad (18)$$

so that $Y(\lambda)$ is the largest y for which $(X(\lambda), y)$ is F -admissible, and ψ is the slack in (B2) for this maximal choice; when X is the extremal value $(1 - M)p^{1/(1-M)}$, $p = 1 - e^{-F'}$, of Theorem 4.2,

(18) is the slack of the self-consistent system for given F and M . Admissibility of a pair (x, y) splits into the two orientations

$$(a) \quad -\log x - s \log y \geq F(s) \quad (0 < s \leq 1), \quad (b) \quad -\log y - s \log x \geq F(s) \quad (0 < s \leq 1),$$

(b) being the condition at ratios $t = 1/s \geq 1$, where $\tilde{F}(t) = tF(1/t)$. Each orientation is certified by one tangent line: if $u \in (0, 1]$ then, by concavity, $F(s) \leq F(u) + (s - u)F'(u)$ for all $s \in (0, 1]$, and the inequality $F(u) + (s - u)F'(u) \leq -\log x - s \log y$ is *linear* in s , so its validity at $s = 0$ and $s = 1$ implies (a); this is Lemma 5.1 with $\varphi = F$, witness $t_0 = u$ and $\sigma = F'(u)$. Likewise for (b), with x and y interchanged. In contrast with the closed-form ansatz of the previous revision of this paper, no concavity of $\tilde{F}$ across the kink at $t = 1$ is required: the two orientations are certified separately, each inside $(0, 1]$.

5.2 The spline certificate

The proof object replaces the explicit functions of the previous revision by *free data*: a list of $N = 35\,770$ cells covering $[\lambda_0, 1]$, $\lambda_0 = 10^{-4}$ (a subdivision of the 10 560-interval partition on which the fixed point was computed; Section 5.3). Cell j ($0 \leq j < N$) carries two *nodes* $(\lambda_j, F_j, d_j, M_j)$ and $(\lambda_{j+1}, F_{j+1}, d_{j+1}, M_{j+1})$ —consecutive cells share a node—two constants X_j, Y_j , and two tangent *witnesses* w_j^a, w_j^b , each of which is literally one of the $N + 1$ nodes of the certificate. All entries are exact scaled integers: λ, d, M, X, Y are integer multiples of 10^{-12} and F of $\frac{1}{2} \cdot 10^{-24}$, so that the trapezoid identity

$$F_{j+1} = F_j + \frac{1}{2} (d_j + d_{j+1})(\lambda_{j+1} - \lambda_j) \quad (19)$$

is *exact* integer arithmetic (the factor $\frac{1}{2}$ is absorbed by the scale of F).

From the data define D and M as the piecewise-linear interpolants of (λ_j, d_j) and (λ_j, M_j) , clamped to the constants d_0, M_0 on $(0, \lambda_0]$; let

$$F(r) = F_0 + \int_{\lambda_0}^r D(t) dt,$$

and let X, Y be the step functions with values X_j, Y_j on cell j , frozen at X_0, Y_0 on $(0, \lambda_0]$. Then F is C^1 and piecewise quadratic with $F' = D > 0$, concave because D is non-increasing, linear on the tail $(0, \lambda_0]$, and $F(\lambda_j) = F_j$ *exactly*, by (19).

Proposition 5.2 (the certificate). *The file `pq_cert_final.json` identified in Section 6 consists of 35 770 cells as above, with $\lambda_0 = 10^{-4}$ and $\lambda_N = 1$. For every cell j , with $\mathcal{D}_j = [d_{j+1}, d_j]$ and $\mathcal{M}_j = [M_j, M_{j+1}]$:*

(C0) *structure*: $0 < d_{j+1} \leq d_j$; $0 < M_j \leq M_{j+1} < 1$; $0 < X_j, Y_j < 1$; and the chain identity (19);

(C1) *the X-inequality*: $X_j \leq (1 - e^{-D})^{1/(1-M)}(1 - M)$ for all $(D, M) \in \mathcal{D}_j \times \mathcal{M}_j$;

(C2) *admissibility*: the witness w_j^a is a node u with (its node values $F(u), D(u)$ being exact data)

$$F(u) - u D(u) \leq -\log X_j \quad \text{and} \quad F(u) + (1 - u) D(u) \leq -\log X_j - \log Y_j,$$

and the witness w_j^b satisfies the same two inequalities with X_j and Y_j interchanged;

(C3) *slack*: with $\hat{\psi}_i = F_i + \frac{1}{2}(\log X_j + \lambda_i \log M_i + \lambda_i \log Y_j)$ for $i \in \{j, j+1\}$ and the dip bound $\delta_j = \frac{1}{8}(\lambda_{j+1} - \lambda_j)(\log M_{j+1} - \log M_j) \geq 0$: $\hat{\psi}_j > \delta_j$ and $\hat{\psi}_{j+1} > \delta_j$.

Moreover the witnesses w_j^a are non-decreasing and the w_j^b non-increasing in j , each one a node of the certificate; and the single tail inequality

$$(T) \quad F_0 - d_0 \lambda_0 + \frac{1}{2} \log X_0 > 0$$

holds. Finally $F_N = 2653604512524788171208898/(2 \cdot 10^{24})$ and $e^{F_N} \leq 3.7689719$.

Proof. By computation, in interval arithmetic at scale 10^{-12} with exp and log enclosed by Taylor series with directed rounding, every comparison being between integers: once by the Lean kernel and once by a bit-exact Python mirror of the Lean checker (Section 5.3). $\square$

Proposition 5.2 implies Theorem 1.1, through Theorem 4.2 in the form of Remark 4.3. First the global hypotheses: $F' = D > 0$; $F(0^+) = F_0 - d_0 \lambda_0 > 0$ by (T), as $\log X_0 < 0$, so $F > 0$; and $C_F < \infty$ because $(H(\lambda) - F(\lambda))/\lambda \leq \log(1/\lambda) + 1 - F(0^+)/\lambda \rightarrow -\infty$ as $\lambda \downarrow 0$. Now fix r in cell j . The interpolants satisfy $D(r) \in \mathcal{D}_j$ and $M(r) \in \mathcal{M}_j$, so (C1) gives (15) for the step value X_j . For (B1): by concavity, $F(s) \leq F(u) + (s - u)D(u)$ for every node u and all $s \in (0, 1]$, so the two endpoint checks (C2), linear interpolation in s , and the tangent bound give, for all $s \in (0, 1]$,

$$F(s) \leq -\log X_j - s \log Y_j \quad \text{and} \quad F(s) \leq -\log Y_j - s \log X_j,$$

i.e. $-\log X_j - t \log Y_j \geq \tilde{F}(t)$ for $t \leq 1$ (first inequality at $s = t$) and for $t \geq 1$ (second inequality at $s = 1/t$, multiplied by t): the pair (X_j, Y_j) is F -admissible. For (B2), write $r = (1 - \theta)\lambda_j + \theta\lambda_{j+1}$. Concavity places F above its chord, $F(r) \geq (1 - \theta)F_j + \theta F_{j+1}$; concavity of log places $\log M(r)$ —the argument being the *linear* interpolant of the M_i —above the linear interpolant of the $\log M_i$; and the convex quadratic $r \cdot$ (linear interpolant of the $\log M_i$) lies below the linear interpolant of the $\lambda_i \log M_i$ by exactly $\theta(1 - \theta)(\lambda_{j+1} - \lambda_j)(\log M_{j+1} - \log M_j)$. Hence

$$\psi(r) \geq (1 - \theta)\hat{\psi}_j + \theta\hat{\psi}_{j+1} - \frac{1}{2}\theta(1 - \theta)(\lambda_{j+1} - \lambda_j)(\log M_{j+1} - \log M_j) \geq \min(\hat{\psi}_j, \hat{\psi}_{j+1}) - \delta_j > 0$$

by (C3), since $\theta(1 - \theta) \leq \frac{1}{4}$. This gives (B2) on $[\lambda_0, 1]$. Finally the tail $r \in (0, \lambda_0]$: there D, M, X, Y are frozen at the values d_0, M_0, X_0, Y_0 , which lie in cell 0's boxes, so (15) holds by cell 0's check (C1), and admissibility of (X_0, Y_0) is a global statement already certified by cell 0's (C2); moreover ψ is *linear* on the tail, equal to $\hat{\psi}_0 > 0$ at λ_0 and to $F(0^+) + \frac{1}{2} \log X_0 = F_0 - d_0 \lambda_0 + \frac{1}{2} \log X_0 > 0$ in the limit $\lambda \downarrow 0$, by (T). Theorem 4.2 now gives validity of F , and $F(1) = F_N$ with $e^{F_N} \leq 3.7689719 < 3.769$ is Theorem 1.1.

This is where self-consistency pays a second dividend. The previous revision needed a hand analysis of small λ (its Lemma 5.3) because its F was pinned to a closed form whose behaviour as $\lambda \downarrow 0$ had to be estimated; the certificate's F is free data and may simply be *continued by a straight line* below λ_0 , self-consistency making the frozen pair (X_0, Y_0) admissible against the extended F itself, with no appeal to the Erdős–Szekeres bound below λ_0 . The one-line inequality (T) is all that remains of the small- λ regime.

5.3 The computer-assisted part

Three computations occur in this paper, with different status.

(a) A *heuristic* floating-point computation (`opt.py`, `opt2.py`) discovers the data; it plays no role in the proof. It computes the fixed point of the bootstrap in the Theorem-13 discretisation of [4]: on each interval of a partition of $[\lambda_0, 1]$, greedily from the left, the minimal slope d admissible for the current F —tested in *both* orientations, self-consistently, against $U = F + \eta$ —together with a per-interval book parameter M and pair (X, Y) ; the sweep is iterated to convergence. The mesh sequence 600, 1200, 2400, 4800, 9600 gives $e^{F(1)} = 3.77003, 3.76943, 3.76912, 3.76897, 3.768894$: clean first-order convergence, with Richardson limit ≈ 3.76881 , so little is left on the table within this framework.

(b) The *proof object* is the exact scaled-integer certificate of Proposition 5.2, exported by `pq_export.py` from the converged fixed point on the 10 560-interval partition: nodal slopes and M are rounded to the integer scales; the F_j are recomputed by the exact chain (19); the X_j, Y_j are rounded down; cells are subdivided (to the final 35 770) until every interval-arithmetic check passes with margin (admissibility margin $6 \cdot 10^{-7}$ per cell, slack margin 10^{-6}); and the base value F_0 is raised by $2.08 \cdot 10^{-5}$ so that the tail inequality (T) holds. The certified constant exceeds the raw fixed point by $7.8 \cdot 10^{-5}$ in total—the price of rounding, of freezing (X, Y) per cell, and of the margins.

(c) An *independent check*: `verify_pq.py`, a bit-exact Python mirror of the Lean checker (on top of `fpi.py`, an op-for-op port of the Lean fixed-point interval arithmetic), re-verifies the chain, every cell, the witness merge and the tail in about two minutes. The strongest check, however, is the Lean kernel itself (Remark 4.4): the formal development is self-contained and trusts none of the Python computations.

The certified terminal value is the exact rational $F_N = 2653604512524788171208898/(2 \cdot 10^{24}) = 1.32680225\dots$, and the kernel-checked enclosure (exp of the halved argument by a degree-16 Taylor series with directed rounding, then squared) gives $e^{F_N} \leq 3.7689719 < 3.769$.

6 Both orientations, and two remarks

The first version of [4]

The first arXiv version of [4] (v1, 26 July 2024) derives the admissible pairs from Lemma 14 there: if $R(k, \ell) \leq e^{\phi(\ell/k)k+o(k)}$ for all $\ell \leq k$, with $\phi(\lambda) = H(\lambda) - \alpha\lambda$, then $(x, e^\alpha(1-x)) \in \mathcal{R}$ for $0 < x < 1/2$. The proof first shows $R(k, \ell) \leq x^{-k-o(k)}y^{-\ell}$, $y = e^\alpha(1-x)$, for $\ell \leq k$, and then asserts that for $x \leq y$ “ $x^{-k-o(k)}y^{-\ell} \leq x^{-\ell}y^{-k-o(k)}$ ” in order to cover the pairs with $\ell > k$. This inequality is reversed: for $x < y$ and $\ell < k$ one has $x^{-k}y^{-\ell} = x^{-\ell}y^{-k}(y/x)^{k-\ell} > x^{-\ell}y^{-k}$. What the hypothesis gives for $\ell > k$ is $R(k, \ell) \leq \binom{k+\ell}{\ell}e^{-\alpha k}$, whereas $x^{-k}y^{-\ell}$ demands $e^{-\alpha \ell}$; the two differ by $e^{-\alpha(\ell-k)}$. Numerically, in the last iteration of v1 (their F_3 with $\beta_3 = 0.03$ and $\alpha_3 = (0.17 - 0.033)/e$, at $\lambda = 1$, where $(X, Y) = (0.23413, 0.80546)$), one finds

$$\min_{t>0} [-\log X - t \log Y - \tilde{F}_2(t)] = -0.00115 \quad \text{at } t \approx 3.8,$$

with F_2 the previously established bound: the pair is *not* F_2 -admissible, the violation occurring exactly in the orientation $\ell \approx 3.8k$. The second version (v2, 29 August 2026) acknowledges that the numerical derivation “was incorrectly justified in the earliest public version” and replaces Lemma 14 by Lemma 15, which handles both orientations (and coincides with Lemma 5.1 above); it also reports a Lean 4 formalisation of the main results. We add that the corrected framework confirms the constant of v1 directly: with $F_3 = H + G$ and $M = \lambda e^{-\lambda}$, the self-consistent slack (18) satisfies $\psi(1) = 6.1 \cdot 10^{-4} > 0$ and, numerically, $\psi > 0$ on $(0, 1]$ (with

$Y(1) = 0.80534$ instead of 0.80546), so Theorem 4.2 re-proves (1) without any chain of iterations. In the certificate of Section 5 the point is structural: every cell carries one tangent witness *per orientation*, so both orientations are tested for every cell.

A proposed quintic correction

Appendix A.2 of [5] reports the function $F = H + e^{-\lambda}(-0.25\lambda + 0.033\lambda^2 + 0.08\lambda^3 - 0.0778\lambda^5)$, which would give $e^{F(1)} \approx 3.6961$, obtained by an automated search within the framework of [4] v1. For this F , with Y the maximal F -admissible choice (18) and M optimised pointwise, one finds $\psi(1) = -0.094$ and $\psi < 0$ on $[0.6, 1]$: the function is not self-consistent, and no round of the bootstrap whose input bound dominates F pointwise (in particular none starting from (1)) can produce it. This is consistent with the expectation in [4, Remark 17] that constants below 3.75 would require new ideas.

Possible improvements

Within the present framework the remaining gap is pure discretisation: the mesh sequence of Section 5.3 converges at first order to ≈ 3.76881 , about 10^{-4} below the certified constant, so little is left. It is worth recording that the two techniques of this paper subsume each other’s gains. Imposing the self-consistent linear tail on the discretised fixed point changes its value by less than 10^{-8} —the greedy slope recursion is strongly attracting in the base value. And restricting the admissibility requirement to the ratios $t \geq \lambda$ actually consumed by (H1) (the pairs (k', ℓ) needed there have $k' \leq k$) changes nothing: the binding ratio is $\approx 3.8\lambda$ throughout. Genuinely better constants would have to come from the combinatorial part—e.g. a sharper key lemma.

Code and data

All code is at github.com/wamlat/RamseyLean-bootstrap; the previous (3.772) revision is archived at DOI 10.5281/zenodo.22263824, and the present revision is on **master**, tagged **v2.0-ramsey-3.769** (commit of record 23572c0), identified by the SHA-256 hashes below. The Python verifier uses only the standard library. Reproduction:

```
python3 verify_pq.py pq_cert_final.json    bit-exact mirror of the Lean checker, about 2 min
lake build RamseyLean.Bootstrap2.CertMain  kernel verification, about six core-hours
```

(the latter preceded by `lake exe cache get` and followed by `#print axioms` on the three final theorems of Remark 4.4). The kernel build checks the 559 chunk files of `Bootstrap2/CertData2` (64 cells each; about 1.5 GB per checking process), and the chunk targets parallelise up to the memory budget; the run recorded in `verification/` took 27 minutes on 16 x86-64 cores.

<code>pq_export.py</code>	0822413e2f8d99dd53ab0ac016ccffce46955068f182cd7c598e9c2e6862d309
<code>verify_pq.py</code>	362731555535c8275aaffa9dafd3b08fa6d104c7e39f7ee176c4f6023ca83c22
<code>fpi.py</code>	b01c4d07ac271ea74e104c652c639b04e64a99e4729a5b4bd8b1a4851f839f1f
<code>funcs.py</code>	237344ad8502f404573102446d90694695fd4f0e9eb5fdbfb00095ccdd2cf6ef
<code>pq_cert_final.json</code>	462cc882a82b08711347cb2af526ea54a46290f6088d1bf82cce66159d66ee1d
<code>opt.py</code>	0a91be491c4ac87d988c94f784cf27cc988fa9a8edff3ec858b55017690093ff
<code>opt2.py</code>	168fe19d9eb5c6e478a7d2e1b095c1eeaa991620753aed2ecefcb2d9e431b5caa

The Lean 4 formalisation (Remark 4.4) consists of the shift ladder (`SelfConsistent.lean`, unchanged from the previous revision) and the directory `RamseyLean/Bootstrap2/`: the certificate

data types and spline definitions, the analytic layer (F from the data by integration, tangent bound, interpolant boxes), the cell checker with its soundness proof, the coverage, witness-merge and tail passes, the glue, and the 35 770-cell certificate data in 559 chunk files.

SelfConsistent.lean	e4b9e7780fc11ecac9e7f516e6e5f074abd60f884a1434269a1baee08b4a6143
Bootstrap2/Defs.lean	e1999435791780a08e655a0436de2858f329a45d9aef5348d9d2fece0ad6e1cb
Bootstrap2/Spline.lean	95017cf347a0e6a969923343187a3dc2a61c370825846ce2acdd573b0f087132
Bootstrap2/Check.lean	a954f59aa2da3cdb88c19c82435b4f252942ef8fc03f847824fe69ce04d4d43f
Bootstrap2/Cover.lean	28eaadb92c83b973816b1526c90ef6fa5e0477342882dbb021885d85bf0d12fe
Bootstrap2/Glue.lean	440000987f5cb4a934fec48eef146041b56295d31efa024391ee6ff5b9cc93f5
Bootstrap2/Main2.lean	7b709fbca384b94c92b8426909f4b119a31e0a81712ade5ac5e3891f0f21e048
Bootstrap2/CertMain.lean	20fd323b9ccfb7f1108bbd9c5e41c067291b7b700a92edceb697d7b76e5e9c74
Bootstrap2/CertData2 (cat Chunk001..559 CertData2.lean)	2ee95b1bc491019c77ec9a4462a8446e76ae64456769948717f61531406d72fa

The repository also retains, unchanged, the artefacts of the previous revision (the Arb [1] certificate, its two checkers, and the directory `lean/Bootstrap/`), and the scripts reproducing the numbers quoted in Section 6.

References

- [1] F. Johansson, *Arb: efficient arbitrary-precision midpoint-radius interval arithmetic*, IEEE Trans. Comput. 66 (2017), 1281–1292.
- [2] M. Campos, S. Griffiths, R. Morris, J. Sahasrabudhe, *An exponential improvement for diagonal Ramsey*, Ann. of Math. (2) 203 (2026); arXiv:2303.09521.
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- [4] P. Gupta, N. Ndiaye, S. Norin, L. Wei, *Optimizing the CGMS upper bound on Ramsey numbers*, arXiv:2407.19026 (v1: 26 July 2024; v2: 29 August 2026).
- [5] *HorizonMath: Measuring AI progress toward mathematical discovery*, arXiv:2603.15617v1 (2026), Appendix A.2.
- [6] N. Ndiaye, *Guarantee of cliques and uncertainty of trees: Ramsey theory and leaf powers*, PhD thesis, McGill University, 2025.